McDiarmid's Inequality for Local Stochastic Noise

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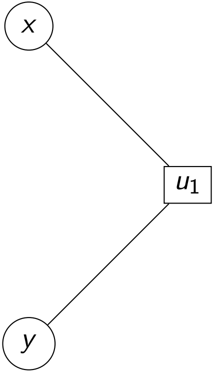
Introduction

Today:

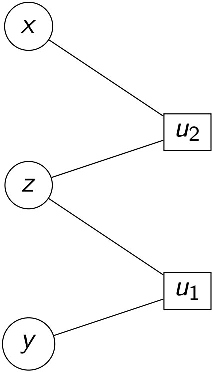
- McDiarmid's Inequality for Local Stochastic Noise.
- Dependence.
- Local Stochastic Noise.
- IQP.

Dependence Type I



$$\begin{aligned} &\mathbf{Pr}\left[(x, y) \in e \middle| u_1 \text{ is satisfied}\right] \\ &= \mathbf{Pr}\left[x \in e \middle| u_1 \text{ is satisfied}\right] \mathbf{Pr}\left[y \in e \middle| u_1 \text{ is satisfied}\right] \\ &\leq \mathbf{Pr}\left[x \in e\right] \mathbf{Pr}\left[y \in e\right] \end{aligned}$$

Dependence Type II



$$
\begin{aligned}
&\mathbf{Pr}\left[(x, y) \in e | z \text{ is not flipped}\right] \\
&= \mathbf{Pr}\left[x \in e | z \text{ is not flipped}\right] \mathbf{Pr}\left[y \in e | z \text{ is not flipped}\right] \\
&\leq \mathbf{Pr}\left[x \in e\right] \mathbf{Pr}\left[y \in e\right]
\end{aligned}
$$

Definition

Consider two (qu)bits x, y. We say that z is a **linked node** between x and y if z is either a bit or a check that is shared by both of their D light cones.

Remark

Notice that y might be a **linked node** between x and y.

Definition

Consider an application of a noise channel. We say that z is a **clean linked node** if:

1. z is a bit and $z \notin f$ (not flipped).
2. z is a check and z is satisfied.

$$
\begin{aligned}
&\mathbf{Pr}\left[x_1, x_2, ..x_k, x_{k+1}, ..x_l \in e | z_1, z_2, ..z_m\right] \\
=&\prod_i \mathbf{Pr}\left[x_i \in e | z_1, z_2, ..z_m\right] \\
=&\prod_i \mathbf{Pr}\left[x_i \in e | z \in \Gamma(x_i)\right] \\
\leq&\prod_i^k \mathbf{Pr}\left[x_i \in e\right] \prod_{i=k+1}^{l} \mathbf{Pr}\left[x_i \in e | z_i \in \Gamma(x_i)\right] \\
\leq& M(p)^k
\end{aligned}
$$

McDiarmid's Inequality - Local Stochastic Noise

C_1, C_2- local stochastic noise

Let $C_1, C_2 > 0$. We say that ρ is subjected to C_1, C_2- local stochastic noise, if for any subset of qubits S the probability they will be faulty is bounded by:

$$C_1 p^{|S|} \leq \mathbf{Pr}\left[S \in e\right] \leq C_2 p^{|S|}$$

Lemma: McDiarmid's Inequality

McDiarmid's Inequality is correct for C_1, C_2- local stochastic noise.

Proof

Proof

Consider ref. We have to bound:

$$\star = \sup_{x,\bar{x}} \mathbf{E}\left[f | X_1, ..X_{i-1}, x\right] - \mathbf{E}\left[f | X_1, ..X_{i-1}, \bar{x}\right]$$

First we show that:

$$\begin{aligned}
\frac{C_2}{C_1} p^{|Z|} &\leq \mathbf{Pr}\left[z_{i+1}, ..z_n | X_1, ..X_{i-1}, x\right] \leq \frac{C_2}{C_1} p^{|Z|} \\
\frac{C_2}{C_1} p^{|Z|} &\leq \mathbf{Pr}\left[z_{i+1}, ..z_n | X_1, ..X_{i-1}, \bar{x}\right] \leq \frac{C_2}{C_1} p^{|Z|}
\end{aligned}$$

Proof

And then we get that:

$$\begin{aligned}
\star &\leq \sum_{Z} p^{|Z|} \left(\frac{C_2}{C_1} - \frac{C_1}{C_2} \right) c \\
&\leq c \left(\frac{C_2}{C_1} - \frac{C_1}{C_2} \right) \frac{1}{1-p}
\end{aligned}$$

Trick.

Suppose that $\mathbf{Pr}[S]$ is not bounded from beneath by $C_1 p^{|S|}$. Then one can define the distribution $\mathcal{D}'$ such that:

$$\mathcal{D}'(S) = \begin{cases} \mathcal{D}(S) & \text{if } \mathcal{D}(S) > C_1 p^{|S|} \\ C_1 p^{|S|} & \text{else} \end{cases}$$

Normalize by:[1]

$$\begin{aligned} \xi &= \sum_{\mathbf{Pr}\left[\mathcal{D}(S) < C_1 p^{|S|}\right]} C_1 p^{|S|} + \mathbf{Pr}\left[\mathcal{D}(S) > C_1 p^{|S|}\right] \\ &\geq \begin{cases} \sum_S p^{|S|} \geq 1 & \text{if } C_1 < 1 \\ C_1 \sum_S p^{|S|} \geq C_1 & \text{else} \end{cases} \\ &\geq C_1 \end{aligned}$$

[1] if $C_1 < 1$

Trick.

So for $C_1 > 1$ we have:

$$\begin{aligned}\mathbf{E}_{\sim\mathcal{D}'}[f] =& \frac{1}{\xi}\sum_{\mathbf{Pr}\left[\mathcal{D}(S)<C_1p^{|S|}\right]} f(S)C_1p^{|S|} + \sum_{\mathbf{Pr}\left[\mathcal{D}(S)\geq C_1p^{|S|}\right]} f(S)D(S)\\ \leq& \frac{1}{\xi}\mathbf{E}_{\sim\mathcal{D}}[f]\\ \leq& \mathbf{E}_{\sim\mathcal{D}}[f]\end{aligned}$$

Trick.

Then:

$$
\begin{aligned}
&\mathbf{Pr}_{\sim\mathcal{D}}\left[f - \mathbf{E}_{\sim\mathcal{D}}\left[f\right] > t\right] \\
&\leq \mathbf{Pr}_{\sim\mathcal{D}'\text{not normalized}}\left[f - \mathbf{E}_{\sim\mathcal{D}}\left[f\right] > t\right] \\
&\leq \frac{1}{C_1}\mathbf{Pr}_{\sim\mathcal{D}'}\left[f - \mathbf{E}_{\sim\mathcal{D}}\left[f\right] > t\right] \\
&\leq \mathbf{Pr}_{\sim\mathcal{D}'}\left[f - \mathbf{E}_{\sim\mathcal{D}}\left[f\right] > t\right] \\
&\leq \mathbf{Pr}_{\sim\mathcal{D}'}\left[f - \mathbf{E}_{\sim\mathcal{D}'}\left[f\right] > t\right]
\end{aligned}
$$

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M : (0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift-rule and the flip upon majority are local decoders.

Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.
For a subset of qubits S, we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)